

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. ATKT Examination (CL)

November 2013

CL305 Environmental Engineering-II

Date: 25.11.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 Do as required:

[07]

1. A pipe or conduit used to convey sewage is called _____.
2. BOD of sewage is generally less than its COD. Why?
3. Corrosion in sewers occurs at:
(A) bottom (B) crown (C) side walls (D) part of sewer that is under water
4. Self-cleansing velocity in sewers >600 mm diameter should be:
(A) 0.1 m/s (B) 0.6 m/s (C) 2 m/s (D) none of these
5. Match correct pairs, Column 1(Treatment process): Column 2 (Level of treatment)

Activated sludge process	Primary treatment
Bar screen	Secondary treatment
Primary sedimentation	Preliminary treatment

6. Match correct pairs, Column 1(waste): Column 2 (origin)

Sewage	Bottom of sedimentation tank
Sullage	Latrine
Sludge	Kitchen

7. MnSO_4 and alkali-iodide reagents are added to a BOD bottle containing water sample to determine dissolved oxygen (DO). If brown precipitates are formed, it indicates:
(A) Absence of DO (B) presence of DO (C) Nothing can be said

Q-2 (a) Explain in brief population equivalent.

[04]

- (b) A 5 mL sample was added to 300 mL BOD bottle containing dilution water. If the initial DO was 8.3 mg/L and final DO at the end of 5-day incubation period was 3.1 mg/L, calculate 5-day BOD of the sample. If the rate constant K (base 10) $= 0.2 \text{ day}^{-1}$, calculate ultimate BOD of the sample.

[06]

- (c) Compute velocity through a coarse screen when approach velocity is 0.6 m/s and the measured head loss is 40 mm.

[04]

OR

Q-2 (a) Which type of settling prevails in a primary sedimentation tank used for sewage treatment? Design a circular primary sedimentation tank for a population of 80000. The detention period is 2.5 h and surface overflow rate is $40 \text{ m}^3/\text{m}^2\text{-d}$. Average sewage generation rate may be assumed as 120 L/capita-day.

[07]

- (b) A circular sewer 60 cm in diameter is laid at a gradient of 1 in 200. Using $N=0.0125$, calculate velocity and discharge if the sewer is running full.

[05]

- (c) Differentiate between self cleansing velocity and non-scouring velocity in sewers.

[02]

Q-3 (a) Explain with a neat sketch biological process occurring in a trickling filter

[06]

- (b) Determine size of a trickling filter for the following data:

[06]

Flow = $1500 \text{ m}^3/\text{d}$, recirculation ratio = 1.2, BOD of sewage at inlet to trickling filter = 200 mg/L, treated effluent BOD desired = 30 mg/L. Assume depth of media = 1.8 m

- (C) Differentiate between surface aeration and diffused aeration [02]

OR

- Q-3 (a) Differentiate between attached and suspended growth processes [05]
 (b) Briefly describe various limitations of BOD test. [05]
 (c) An activated sludge unit has been designed to treat sewage flow of 10000 cu.m/day and inlet COD of 350 mg/L. The MLVSS concentration in aeration tank is 2500 mg/L. The dimensions of aeration tank are: 60 m (L) x 15 m (W) x 4 m (liquid depth). Calculate retention time and F/M ratio in aeration tank. [04]

SECTION - II

- Q-4 1. The primary air pollutant which is formed due to incomplete combustion of organic matter is: [07]
 (A) methane (B) sulfur dioxide (C) ozone (D) carbon monoxide
 2. The secondary air pollutant among the following is:
 (A) hydrocarbons (B) sulfur dioxide (C) ozone (D) carbon dioxide
 3. Photochemical smog is:
 (A) smoke + mog (B) smart + fog (C) Smoke + fog (D) small + fog
 4. Which of the following air pollutants reduces oxygen carrying capacity of blood and may cause death by asphyxiation?
 (A) CO_2 (B) SO_2 (C) hydrocarbons (D) CO
 5. important air pollutants responsible for acid rain are:
 (A) CO_2 & H_2S (B) CO & NO (C) SO_2 & NO_x (D) NO_x & O_3
 6. Electrostatic precipitator is most useful in one of the following industries:
 (A) textile (B) hydroelectric power plant (C) thermal power plant (D) dairy
 7. Which one of the following methods of solid waste disposal is the most environment-friendly?
 (A) landfilling (B) incineration (C) composting (D) open dumping

- Q-5 (a) Explain various kinds of plant leaf damages due to air pollutants. [06]
 (b) Differentiate between garbage and rubbish [02]
 (c) Describe principle, construction & working of a cyclone with neat sketch. [06]

OR

- Q-5 (a) Describe with neat sketch the principle, construction & working of a Fabric filter Precipitator. [06]
 (b) Describe with neat sketch disposal of solid waste in a sanitary landfill [06]
 (c) Define any one: 1. Primary air pollutants, 2. Secondary air pollutants [02]

- Q-6 (a) Differentiate between composting & vermicomposting. [03]
 (b) Briefly mention advantages and disadvantages of environmental auditing. [05]
 (c) What are the general applications of the following air pollution control equipments? [06]
 (i) Electrostatic precipitator, (ii) packed bed adsorber, (iii) ventury scrubber

OR

- Q-6 (a) Describe various factors affecting quantity of municipal solid waste. [06]
 (b) Distinguish between Dry & Wet Precipitators. [03]
 (c) Write a short note (any one): [05]
 (1) Leachate from landfill site, (2) Indore method of refuse disposal, (3) collection of municipal solid waste.

List of useful formulae

$$E = \frac{100}{1 + 0.4432 \sqrt{\frac{W1}{VF}}}$$

$$F = \frac{1 + r}{(1 + 0.1r)^2}$$

$$v = \frac{1}{n} r^{\frac{2}{3}} S^{\frac{1}{2}}$$

$$H_L = 0.0729(V^2 - v^2)$$

$$V_s = \sqrt{\frac{8\beta g}{f}(S_s - 1)d}$$

$$V_s = 60.6(S_s - 1)d(3t + 70)/100$$

$$F/M = S_o/X_v \cdot t$$

$$SOR = \text{Flow/surface area}$$

$$Y_t = L_o(1 - 10^{-kt})$$

$$k_T = k_{20}(1.047)^{(T-20)}$$

$$BOD = (D_i - D_o)/P$$

$$L = L_o \cdot 10^{-kt}$$